Secure Your Network with Snort

This step-by-step installation guide will get you familiar with Snort, a popular intrusion detection system.

Snort is an open source network intrusion detection system (NIDS) that is widely used for monitoring network traffic in real-time and analysing packets for signs of intrusion and malicious activity. Developed by Martin Roesch in 1998, Snort has evolved into a robust security tool that is essential for network administrators and security professionals.

Snort operates by capturing network traffic and analysing the content of each packet against a database of signatures or rules designed to identify malicious behaviour. This process allows Snort to detect a variety of attacks and probes, such as buffer overflows, stealth port scans, CGI attacks, SMB probes, and OS fingerprinting, among others.

How Snort works

Snort inspects packets entering and leaving a network interface and matches them against a database of signatures or known characteristics of malicious threats. When a match is found, Snort acts based on predefined rules. These actions can include logging the packet details, generating alerts, or even actively blocking the traffic.

The system uses three primary modes of operation.

Sniffer mode: In this mode, Snort reads network packets and displays them on the console. This is useful for network traffic debugging.

Packet logger mode: Snort logs packets to the disk. This mode is helpful for network traffic analysis and conducting investigations after an event has occurred.

Network intrusion detection mode: This is the most sophisticated mode where Snort analyses network traffic against a database of known attack signatures and performs actions based on what it detects.

Key features of Snort

Real-time analysis: Snort provides live monitoring of network traffic, allowing for the immediate detection of potential threats.

Customisable detection: With its flexible rule-based language, Snort enables users to write and modify rules to fit their specific environment and security policies.

Protocol analysis: Snort can analyse a wide range of network protocols, making it possible to identify threats at different layers of the network stack.

Content matching: The system can search for specific content patterns within packet payloads to identify complex threats.

Open source community: Snort benefits from a vibrant community that contributes to its rule database, ensuring it remains up-to-date with the latest security threats.